Last week:

- Eulerian trails and circuits
- Hamiltonian paths and cycles

This week:

- Matchings
- Edge- and vertex-covers
- Edge- and vertex- independence and covering numbers $\alpha, \alpha', \beta, \beta'$.

Recall: An independent set of a graph G is a subset A of vertices such that each edge in G has at most one endpoint in A.

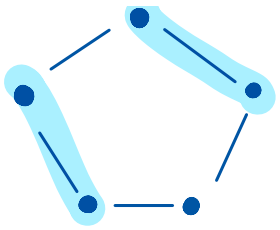
Definition. *Let $G = (V, E)$ be a graph.*

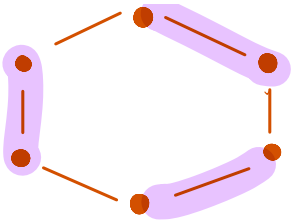
- *A **matching** on G is a subset $M \subset E$ such that each $v \in V$ is an endpoint of at most one edge in M.*
  ***Equivalently:** a matching is a set of edges which are vertex-disjoint.*
- *The **size** of a matching is the number of a edges in it.*
- *Let $A \subset V$. We say that a matching M **saturates** A if every vertex in A is incident to some edge in M.*
- *A matching is **perfect** if it saturates V.*

$|M| \leq \left\lfloor \frac{|V(G)|}{2} \right\rfloor$

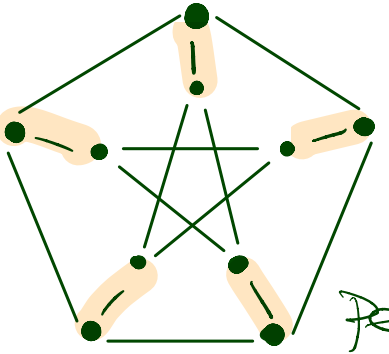
if M is perfect then $|V(G)|$ is even, $|M| = \frac{|V(G)|}{2}$

Example.





Perfect!

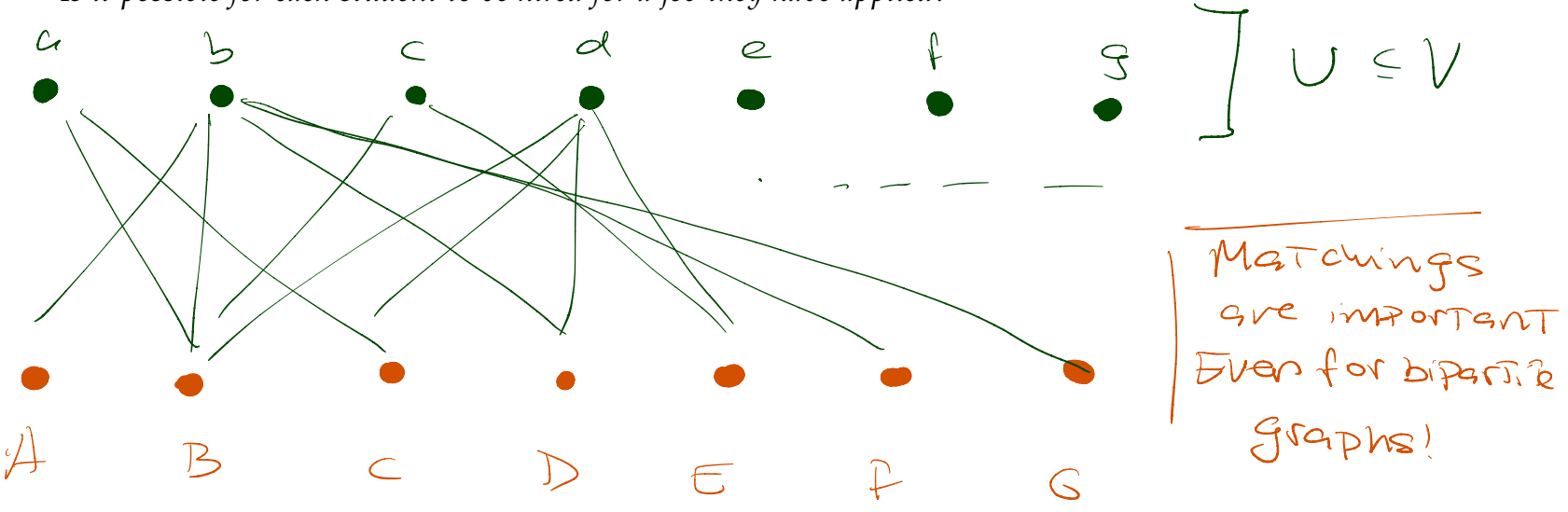


Perfect!

Example. *Seven seniors a, b, c, d, e, f, g apply for positions A, B, C, D, E, F, G according to the following list:*

- *a: B, C*
- *b: A, B, D, F, G*
- *c: B, E*
- *d: B, C, E*
- *e: A, C, D, F*
- *f: C, E*
- *g: D, E, F, G*

Is it possible for each student to be hired for a job they have applied?

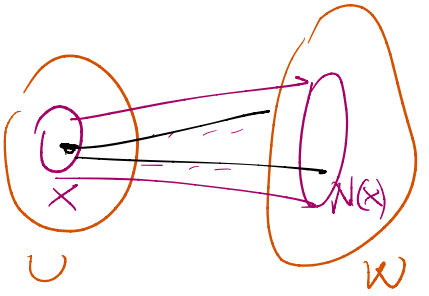


Upshot: Does G contain a matching that saturates U?

Definition. *Let* $G = (V, E)$ *be a graph and let* $X \subset V$. *The* ***neighborhood*** $N(X)$ *of* X *is*

All vertices adjacent to some element in X

$$N(X) = \bigcup_{x \in X} N(x) = \{v \in V : vx \in E \text{ for some } x \in X\}$$



Definition. *Let* G *be a bipartite graph with parts* U, W *such that* $|U| \leqslant |W|$. *We say that* G *satisfies the* ***Hall's condition*** *if*

$$|N(X)| \geqslant |X| \qquad \text{for every non empty } X \subset U.$$

Theorem (8.3 Hall's Theorem). *Let* G *be a bipartite graph with parts* U, W *such that* $|U| \leqslant |W|$. *Then* G *contains a matching which saturates* U *if and only if* G *satisfies Hall's condition.*

Proof: ⇒) "Obvious restriction"

Let $M \subset E$ matching that saturates U.

Now, for any $u \in U$ there is a unique $w_u \in V$ such that $uw_u \in M$

(w_u exists because M saturates U, w_u is unique because M matching

Moreover, since G is bipartite $w_u \in W$

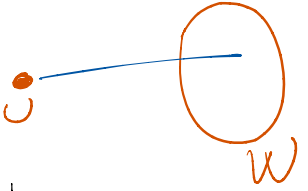
Thus $u \longmapsto w_u$ is an injective function $U \longrightarrow W$

Take any $\emptyset \neq X \subseteq U$, Then f restricted to X is a injective function $X \longrightarrow N(X)$, Hence $|X| \leq |N(X)|$

$\Leftarrow$) "Obvious restriction is unique restriction"

By strong induction on $r = |U|$

Base case $r = 1$



We have $|N(u)| \geq |\{u\}| = 1$, so there is some neighbor w of u

But $w \in W$. Thus $M = \{uw\}$ is a matching that saturates $U = \{u\}$.

IH: Assume $\Leftarrow$) true for any bipartite graph with $|U| < r$

Let G bipartite, $V = U \cup W$, $r = |U| \leq |W|$.

Assume G satisfies Hall's

Want: matching saturating U.

Case 1: Every proper subset X of U satisfies $|N(X)| > |X|$.

Take $u \in U$, Then it has some neighbor $w \in W$

Consider $G' = G - \{u, w\}$, bipartite with parts $U - \{u\}$, $W - \{w\}$

Take $\emptyset \neq X \subseteq U - \{u\}$, since $X \subseteq U$, we have $|N_G(X)| \geq |X|$

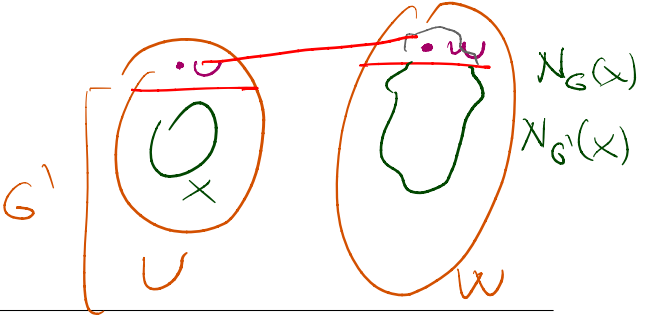
But in $N_G(X)$ we are deleting at most w. $(u \notin X)$

Then $|N_{G'}(X)| \geq |N_G(X)| - 1 > |X| - 1$

Hence $|N_{G'}(X)| \geq |X|$

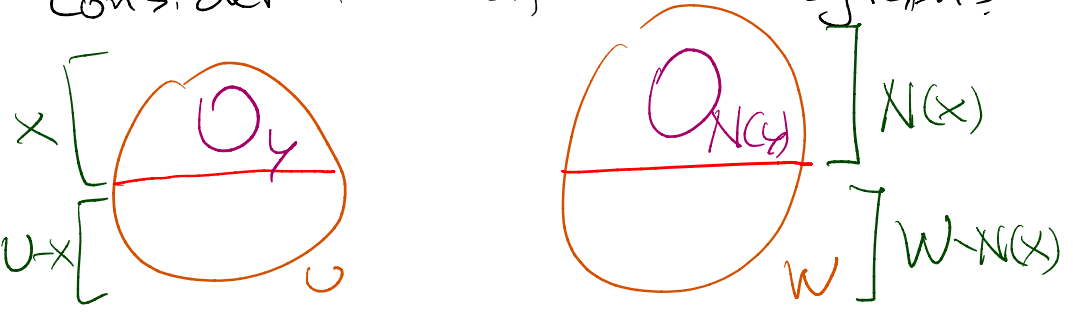
By IH, G' has a matching M' that saturates $U - \{u\}$

Hence $M = M' \cup \{uw\}$ matching saturates U



Case 2: There exists a proper subset X of U such that $|N(X)| = |X|$.

Consider two bipartite subgraphs:



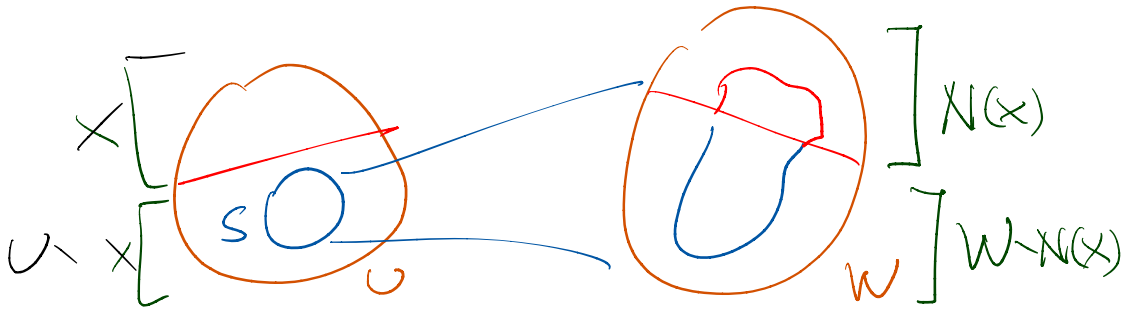
G_1 with parts X, N(X)

G_2 with parts U-X, W-N(X)

Claim 1: G_1 satisfies Hall's.

Indeed if $\emptyset \neq Y \subseteq X$, Then $|Y| \leq |N_G(Y)| = |N_{G_1}(Y)|$

Thus G_1 has a matching M_1 saturates X.



G_2 parts $U \setminus X$, $W \setminus N(X)$

Claim 2: G_2 satisfies Hall's

Take $\emptyset \neq S \subseteq U \setminus X$. Then

$N_{G_2}(S) = N_G(S) \setminus N(X)$, also $N_G(X \cup S) = N(X) \sqcup (N(S) \setminus N(X))$

$|X| + |S| \leq |X \cup S| \leq |N(X \cup S)| = |N(X)| + |N(S) \setminus N(X)|$

↳ Hall's for G $= |X| + |N_{G_2}(S)|$

Hence $|S| \leq |N_{G_2}(S)|$ ✓

By IH, G_2 has a matching M_2 saturating $U \setminus X$

Then $M = M_1 \sqcup M_2$ matching for G saturates U. ∎

Definition. *A* ***system of distinct representatives (SDR)*** *for a collection* $S_1, \dots, S_n$ *of finite non-empty sets is a family of distinct elements* $s_1, \dots, s_n$ *such that* $s_i \in S_i$ *for each* $i \in \{1, \dots, n\}$.

Theorem (8.4, Original formulation of Hall's theorem). *A collection* $S_1, \dots, S_n$ *of finite non-empty sets has a SDR if and only if for each* $k \in \{1, \dots, n\}$, *the union of any* k *of these sets contains at least* k *elements*.

"Proof": We define a graph G with parts

$U = \{v_1, \dots, v_n\}$, $W = \bigcup_{i=1}^{n} S_i$

with an edge $v_i w \in E(G) \iff w \in S_i$

Then an SDR is just a matching that saturates U

And Hall's condition is equivalent to



Theorem (8.6). *For any* $r \geq 1$, *every* r*-regular bipartite graph has a perfect matching.*

Proof: Let U, W denote the parts. Note $|U| = |W|$, because

$$\sum_{u \in U} \deg u = |E(G)| = \sum_{w \in W} \deg w$$

$$\| \qquad\qquad \| \qquad\qquad \| \qquad\qquad \| \qquad\qquad \|$$

$$r|U| \qquad\qquad r|W| \implies |U| = |W|$$

Next, we see has a matching that saturates U.
(since $|U| = |W|$, that matching is perfect)

WTS: G satisfies Hall's. Take $\emptyset \neq X \subseteq U$. Let $k = |X|$

WTS: $|N(X)| \geq k$. There are kr going out of X to $N(X)$

Thus $N(X)$ receives at least kr edges. Each vertex in $N(X)$ takes r of those. Thus $|N(X)| \geq k$. ✓



Definition. *Let* $G = (V, E)$ *be a graph*

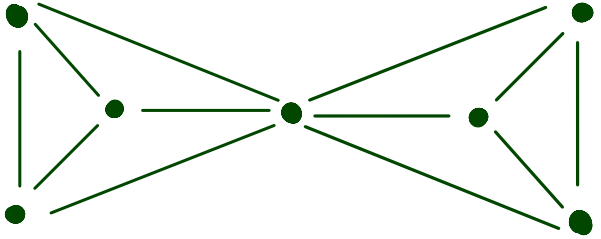
(1) *The* ***independence number*** $\alpha(G)$ *is the size of a largest independent set in* G.

(2) *The* ***edge-independence number*** $\alpha'(G)$ *is the size of a largest matching in* G. $\alpha' \leq \left\lfloor \frac{V(G)}{2} \right\rfloor$

(3) *A* ***vertex-cover*** *of* G *is a subset* $U \subset V$ *such that every edge of* G *has at least one end-point in* U.
The ***vertex-covering number*** $\beta(G)$ *is the size of a smallest vertex-cover of* G.

(4) *An* ***edge-cover*** *of* G *is a subset* $S \subset E$ *such that every vertex of* G *is incident to at least one edge in* S.
The ***edge-covering number*** $\beta'(G)$ *is the size of a smallest edge-cover of* G.

Example.



Note: Let G be a bipartite graph with parts U, W such that $|U| \leqslant |W|$. Then:

- $\alpha'(G) \leqslant |U|$, and
- $\alpha'(G) = |U|$ if and only if G contains a matching which saturates U.

As a consequence, we can rewrite Hall's Theorem:

Theorem (8.3 Hall's Theorem)**.** *Let G be a bipartite graph with parts U, W such that $|U| \leqslant |W|$. Then $\alpha'(G) = |U|$ if and only if G satisfies Hall's condition.*

Exercise. *Consider integers $n \geqslant 3$ and $1 \leqslant s \leqslant t$. Convince yourself of the following:*

- $\alpha(C_n) = \lfloor n/2 \rfloor$, $\alpha(K_n) = 1$, $\alpha(K_{s,t}) = t$.
- $\alpha'(C_n) = \lfloor n/2 \rfloor$, $\alpha'(K_n) = \lfloor n/2 \rfloor$, $\alpha'(K_{s,t}) = s$.
- $\beta(C_n) = \lceil n/2 \rceil$, $\beta(K_n) = n - 1$, $\beta(K_{s,t}) = s$.
- $\beta'(C_n) = \lceil n/2 \rceil$, $\beta'(K_n) = \lceil n/2 \rceil$, $\beta'(K_{s,t}) = t$.

In particular, verify that for any of the graphs G above, $\alpha(G) + \beta(G) = |V(G)|$ and $\alpha'(G) + \beta'(G) = |V(G)|$.